# GCI World 2026 April

# **Final Assignment README**

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| Deadline | 2026 June 19th, 11:00 UTC |
| Submission via | Omnicampus<br>https://edu.omnicamp.us/courses/137/ |

## 1. Scenario & Objective

You are a new associate at an IT consulting firm. The sales team has just secured a PoC (Proof of Concept)* project from a company. The company has a large dataset but has limited in-house expertise to utilize them for business. **Your job is to run EDA and build at least one machine learning model on the dataset. From evidence and market analysis, you need to define the client's core business problem. Then, you will create a data-driven business proposal with quantified impact (e.g. revenue lift, cost savings, risk reduction, etc.)**. As an associate, your proposal must be data-driven and machine learning-based. A strong proposal may lead to a formal consulting contract and give you the opportunity to lead a machine learning–based business development project.

***Proof of Concept (PoC)**: an experimental verification process conducted before fully implementing a new technology or idea. Its purpose is to confirm feasibility and requirements. If successful, the project may advance to full-scale system development or business strategy formulation.

**Output**: Create a presentation (up to 15 slides including title page, references, and any section dividers) that outlines your business model proposal.

## 2. Requirements

- **Your submission will be evaluated based on whether it meets the following requirements**. We encourage you to use the lecture materials as it will help you in satisfying all requirements.
- Double-check each requirement before submission. Failing to meet them may result in a non-passing grade.

1. Move beyond mere technical reporting; form a clear **business proposal** driven by insights from the data.
2. **Base your business proposal on actual EDA of the dataset**, highlighting client-specific challenges.
3. Appropriately build and report at least one machine learning or statistical model
    - **Name the model, evaluation metric, and score**

- **Explain how the model's quantitative insights inform your proposal**

4. Use external sources (data, reports, papers, etc.) and **conduct market analysis**. **Cite sources clearly**.
5. **Make the presentation understandable** to client executives with no ML background.
6. Use multiple **data visualizations**, not just raw numbers.
7. Ensure **logical consistency** from analysis through proposal.

**Notes on requirement (3):**

- Please **explicitly write the model name, evaluation metric, and score in the slide**. Using only visualizations (screenshots, figures, etc.) is not enough.
- If the model used is clearly not appropriate for the data, we will not count as fulfilling the requirement.
- Even though real-world executive presentations might skip technical details, this assignment is designed to test your understanding of the course materials. Be explicit about **why you chose each model, how you built it, and the insights you gained from it**.

**Notes on requirement (4):**

- You will need to submit a list of references. See Section 2-1 for details.

## 2-1. Regarding Citations

You are encouraged to reference publicly available data and methods as in scholarly research, but you must cite sources to avoid plagiarism.

1. Cite all external data, methods, figures, or text. **Lack of citation may be judged as plagiarism**.
2. Even if not verbatim copy/paste, any method or visualization borrowed from a source requires citation.
3. You may use additional open data in your analysis and model building, but cite everything.
4. You may include your own market research; again, cite if referencing external data.

- **Quoted or copied material is not the basis of your grade**; your original analysis and synthesis are.

- As a rule of thumb, aim for **<40% of the final material as direct quotation**.
- Show citations on-slides (brief footnote or reference tag) and list all items in the list of references.

## 3. How to Submit

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| Submission Portal | Omnicampus<br>• https://edu.omnicamp.us/courses/137/reports/106/ |
| Deadline | June 19th 11AM UTC |
| Required Materials | 1. Slides (PDF format)<br>• Up to **15** slides total (including title page, table of contents, section dividers, references, etc.).<br>• Submit as a **PDF** exported from PowerPoint, Keynote, Google Slides, Figma, etc.<br>• Slides beyond 15 may be disregarded for evaluation.<br><br>2. List of References<br>• Enter URLs or literature citations in the specified “References” field in Omnicampus.<br>• **If you used any generative AI, please paste the URL to the chat history.**<br>• Citations must be included in the slides as well<br>• The list must match citations used in your slides.<br><br>3. Notebook<br>• In .ipynb or .py format (or ZIP format if multiple files), containing reproducible analysis. |
| File Naming | Slides and notebook should have your Omnicampus account name as the file name.<br>• **Slides**:[OmnicampusAccountName.pdf]<br>• **Notebook**: [OmnicampusAccountName.ipynb], [OmnicampusAccountName.py], or [OmnicampusAccountName.zip] |
| File Size | • **Slide File Size**: ≤ 10MB<br>• **List of References**: No restrictions<br>• **Notebook**: No restrictions |
| Resubmission | You may resubmit multiple times before the deadline. Only your final submission is evaluated. |
| Late Submissions | No late additions or updates to references/notebooks will be accepted once the deadline has passed. |

## 4. Grading

- Grading results will be announced via Slack after the deadline.
- “**Outstanding Students**” will be chosen based on the overall scores including the final assignment, competition scores, attendance surveys, and homework submissions. Details will also be shared via Slack.

## 5. Dataset

Refer to tutorial.ipynb on how to import data.

## 6. FAQ

- **I was sick/busy and could not submit by the deadline. Is late submission OK?**
  - No, we do not accept any late submissions.
- **I found an error in my slide! Can I resubmit?**
  - Yes, you can resubmit your work if it is before the deadline.
  - Note that only your latest submission will be used for grading.
- **When I resubmit my work, can I just resubmit what I changed?**
  - No, you need to attach all files (slides, list of references, notebook).
- **Can I work with my friends?**
  - You may discuss with other students at your school or on Slack channel, but copying someone else’s work will be considered plagiarism.
- **Can I ask GPT5 to do the assignment?**
  - You are free to use any generative AI to discuss business proposals, create diagrams, etc., but copying the output will not be considered as your own work.
  - Make sure to include the chat URL to the list of references.
- **Do I need to have 15 slides?**
  - No, the number of slides is unrelated to grading.
  - However, make sure to include all contents needed to persuade your client.
- **Can I choose any target variable?**
  - Yes, but make sure to justify the reason from EDA + market context.

- **How many models?**
    - At least one, but comparing multiple models is preferable.
- **Can I use extra publicly available data?**
    - Yes, but make sure to cite it and explain how it informs your proposal.

# 7. Additional Hints

## 7-1. Suggested Workflow

1. **Introduction & Market Analysis**
    - Analyze using external data: market size, growth, risks, etc.
    - *What is the situation of the market (globally/locally)?*
    - *What assumptions can be made about the potential needs?*
2. **EDA**
    - Visualizations
    - *What can we infer from the data?*
    - *What are some early hypotheses about the pain points?*
3. **Problem Definition**
    - Define the business objective and the ML task
    - Specify target variable and rationale.
    - *What ML task best aligns with the objective and the client's pain points?*
4. **Feature Engineering**
    - *What kind of feature engineering best fits the problem/dataset?*
5. **Experiments**
    - Model selection and evaluation
    - Visualize results and insights
    - *What model is best suited for the problem? What metrics best align with the objective?*
    - *What can be inferred from the results?*
6. **Business Proposal**
    - Translate EDA and model findings into a business proposal
    - Quantify impact: revenue lift, cost savings, risk reduction, etc.
        i. Include assumptions if needed
    - *What solution solves the problem?*

    - *How does the solution benefit the client quantitatively?*
7. **Next Steps (optional)**
    - Potential risks
    - Detailed implementation plans
    - etc.

## 7-2. How to Improve Your Slides

- Your audience has no ML background, yet skipping technical details entirely is not advisable for a PoC.
- Translate essential methods and implementations into understandable language; use conceptual diagrams to show how your approach is valid and applicable.
- Be clear in your slide titles and messages; consider adding summaries to ensure your points are communicated effectively.
- Design every chart or graph with purpose: choose colors, legends, fonts, and units carefully; handle outliers and binning to avoid misleading visuals.
- Too many figures can distractive, while too few can obscure important findings; strike a balance.